ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE
School of Basic Sciences
Department of Physics

Impact of an adaptive gravitational softening length on the inner density profile of dark matter halos in collisionless simulations

Bachelor's Thesis

by

Martí Ayuso Maldonado

Supervised by

Dr. Yves Revaz

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Laboratory of Astrophysics

Abstract

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Résumé

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1 Introduction

1.1 The cusp-core problem

Dark matter is believed to constitute approximately 85% of the matter in the universe and plays a crucial role in shaping cosmic structure. Despite decades of observational and theoretical effort, its fundamental nature remains one of the outstanding questions in physics and astronomy. At cosmological scales, dark matter halos form through gravitational collapse and hierarchical assembly, producing characteristic density profiles that encode information about the underlying dark matter model, assembly history, and internal dynamics.

In standard Cold Dark Matter (CDM) cosmology, gravitational N-body simulations predict that dark matter halos develop steep, cuspy density profiles in their central regions, following a power-law slope $\gamma \approx -1.5$ to -2 within a characteristic transition radius. Observationally, however, many dwarf galaxies exhibit much shallower central profiles with slopes closer to $\gamma \approx -0.2$ to -1, a feature referred to as a density core. This discrepancy between the predicted cusps and observed cores, known as the *cusp-core problem*, has motivated decades of investigation into whether the resolution lies in the nature of dark matter, the dynamical history of galaxies, or systematic observational uncertainties (Power et al., 2003; Zhang et al., 2019).

One promising explanation for core formation involves non-adiabatic perturbations to the potential. When supernovae, black hole feedback, or other energetic events suddenly alter the gravitational potential of a galaxy, orbits that were previously in equilibrium can be disrupted. Particles at small radii can then diffuse outward through repeated phase-space mixing, gradually flattening the inner density profile. More subtly, stochastic gravitational fluctuations generated by clumpy substructure or halo asymmetries can also transfer energy and angular momentum to central orbits over long dynamical times, producing a cumulative heating effect (Peñarrubia, 2017, 2019). Such mechanisms appear particularly efficient in ultra-faint dwarf galaxies, which have shallow potential wells and are prone to ongoing tidal interactions.

[PLACEHOLDER FIGURE] Illustration of cusp-like and core-like dark matter density profiles. Left: steep central cusp predicted by CDM simulations. Right: shallower core observed in many dwarf galaxies. The transition occurs at a radius that depends on the halo mass and evolutionary history.

Figure 1.1: **[PLACEHOLDER FIGURE]** Cusp versus core density profiles.

1.2 Motivating question

The persistent inner-density decline observed in controlled, dark-matter-only simulations of relaxed halos presents a puzzle. In such runs, there are no baryonic processes (supernovae, AGN feedback) and no time-dependent external perturbations. Gravity is purely Newtonian, computed with modern adaptive algorithms. Yet the simulated halos still exhibit a gradual flattening of their central profiles over time. This raises a critical question: is this decline a reflection of genuine collisionless dynamics, an artifact of numerical approximations in the gravity solver, or a side effect of the post-processing analysis pipeline?

The question is not merely academic. If the observed density decline is primarily a numerical artifact, then the apparent core in simulations is unreliable and informs little about real galaxy evolution. If it is a genuine dynamical feature, then understanding its mechanism opens a window into collisionless relaxation and the role of global halo coupling.

This thesis focuses on determining which of these possibilities explains the persistent central-density decline observed in a reference SWIFT simulation of a dark matter halo evolved in isolation over 100 Gyr.

[PLACEHOLDER FIGURE] Central density evolution over time in a dark-matter-only simulation. The plot shows a gradual decline of the inner density despite no baryonic feedback or external perturbations. The central question is whether this decline is physical, numerical, or methodological in origin.

Figure 1.2: **[PLACEHOLDER FIGURE]** Central density evolution over 100 Gyr.

1.3 Aims and objectives

The main goal of this work is to systematically investigate the origin of the central-density decline through controlled parameter studies, diagnostic validation, and mechanistic tests. The specific objectives are:

1. **Characterize the baseline behavior**: Establish what the reference simulation produces in terms of density-profile evolution, and verify that the trend is robust against simple post-processing variations (centering method, snapshot noise reduction).
2. **Rule out trivial numerical explanations**: Test whether changes to numerical parameters—gravitational softening length, timestep factor, force-solver accuracy settings—significantly alter the observed decline. If numerical parameters prove insensitive, a more fundamental physical or structural process becomes more likely.
3. **Assess global and local influences**: Investigate whether the inner-density trend reflects purely local collapse-scale physics or depends on the larger-scale halo environment. Shell-based diagnostics will quantify how outer layers couple to the center.
4. **Distinguish mass redistribution from heating**: Use cumulative-mass and angular-momentum diagnostics to determine whether particles are being heated into more eccentric orbits, or whether mass is being systematically transported outward.
5. **Evaluate the robustness of the effect**: Test whether the decline persists for different initial halo structures (cusps versus cores) and different simulation volumes, to establish whether the mechanism is universal or tuning-dependent.

The cumulative outcome will be either a validated mechanism (if the trend survives all tests and correlates with a specific physical process) or a ruled-out category of explanations (if specific numerical or methodological factors prove decisive).

1.4 Thesis outline

The thesis is organized as follows:

Chapter 1: Theoretical Background introduces the collisionless framework governing dark matter halos, defines the cusp-core phenomenology, and reviews the numerical considerations that affect halo simulations. This chapter establishes the theoretical minimum needed to interpret the results. We discuss the collisionless Boltzmann equation, the role of gravitational softening and force-solver accuracy, and the importance of shell-to-shell coupling in global halo dynamics.

Chapter 2: Simulation Setup, Framework and Implementation documents the numerical setup in detail. We describe the SWIFT configuration, initial-condition generation, halo parameter choices, diagnostic definitions (density profiles, cumulative mass, shell contributions), and the centering and snapshot concatenation procedures used in post-processing. This chapter is designed for reproducibility: a reader should be able to replicate the simulations and analysis from the information provided.

Chapter 3: Results and Discussion presents the parameter studies and sensitivity analyses. The chapter progresses through six main tests: (i) the baseline evolution, (ii) analysis pipeline robustness, (iii) box-size sensitivity, (iv) initial-condition dependence (cusp versus core), (v) softening-length variations, and (vi) self-gravity solver parameter sensitivity. Each test is framed as a mechanistic hypothesis to either support or eliminate candidate explanations. Later sections address cumulative mass evolution, shell contributions, hybrid models, and synthesis of the findings.

Chapter 4: Conclusions summarizes which explanations have been ruled out and which remain plausible, discuss the implications for collisionless halo dynamics, and outline suggestions for future work.

The thesis therefore focuses on determining whether the effect is numerical, methodological, or physical in origin, by combining targeted diagnostics with controlled variations of the numerical setup and initial conditions.

2 Theoretical background

This chapter introduces the physical and numerical concepts required to interpret the evolution of the inner dark matter density in ultra-faint dwarf galaxy simulations. The objective is not to review all of modern structure formation, but to establish the minimum framework needed for the analyses in subsequent chapters: collisionless halo dynamics, cusp/core phenomenology, numerical heating, gravitational softening, the gravity solver controls used in SWIFT, and the role of shell-to-shell coupling in mass redistribution.

The chapter is intentionally organized from first principles to practical diagnostics. We begin with the collisionless equations that govern halo evolution, then move to the cusp-core problem in faint systems, and finally discuss numerical ingredients that can mimic or amplify physical effects. This progression is important because the central scientific question of the thesis requires separating genuine dynamical evolution from resolution- and solver-induced artefacts.

2.1 Dark matter halos and density profiles

In the collisionless limit, dark matter is described by a phase-space distribution function $f(\mathbf{x}, \mathbf{v}, t)$ that obeys the collisionless Boltzmann equation, coupled to Poisson's equation for the gravitational potential (Binney and Tremaine, 2008). The key implication is that halo evolution is governed by a smooth mean gravitational field rather than by direct microscopic collisions between particles.

Using standard notation, the collisionless system is governed by

$$\frac{\partial f}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{x}} f - \nabla_{\mathbf{x}} \Phi \cdot \nabla_{\mathbf{v}} f = 0, \tag{2.1}$$

with the potential determined by

$$\nabla^2 \Phi = 4\pi G \rho, \qquad \rho(\mathbf{x}, t) = \int f(\mathbf{x}, \mathbf{v}, t)\, \mathrm{d}^3\mathbf{v}. \tag{2.2}$$

These equations make clear that halo structure is a phase-space problem: the density profile cannot be understood independently of the velocity distribution and orbital content.

In cosmological simulations, dynamically relaxed halos are commonly described by broken power-law density profiles, where the logarithmic slope changes from inner to outer radii. In this language, the central shape of the profile is the critical diagnostic: a steep rise toward small radii is referred to as a *cusp*, while a flattened inner profile is referred to as a *core*. Since this inner region is also the most demanding numerically, any physical interpretation must be tied to explicit convergence and resolution arguments (Power et al., 2003; Zhang et al., 2019).

A commonly used reference model is the NFW-like form, which captures the transition from an inner and outer power-law regime. The associated local slope,

$$\gamma(r) \equiv -\frac{\mathrm{d}\ln\rho}{\mathrm{d}\ln r}, \tag{2.3}$$

is a useful quantity throughout this thesis because it directly tracks whether the central profile evolves toward cusp-like or core-like behaviour.

For this thesis, the practical consequence is straightforward: the measured inner density slope cannot be interpreted in isolation. It must be analyzed jointly with the numerical setup that determines force accuracy, relaxation strength, and time integration quality.

2.2 Cusp-core behaviour in ultra-faint dwarf galaxies

Ultra-faint dwarf galaxies are particularly valuable for small-scale dark matter studies because they are strongly dark-matter dominated and have shallow gravitational potentials. Their internal structure is therefore sensitive both to the underlying dark matter model and to dynamical heating processes that can redistribute mass over time (Salamin, 2026).

In a cold dark matter scenario, cuspy inner profiles are the usual expectation for collisionless halos. In warmer dark matter scenarios, free-streaming suppresses small-scale power and can alter assembly histories, potentially affecting central structure indirectly through different accretion and merger pathways (Salamin, 2026). Independently of the particle model, repeated perturbations and non-adiabatic potential fluctuations can transfer energy to central orbits and drive profile flattening.

The conceptual point is that core formation is not a binary signature of one mechanism. Similar-looking inner profiles can emerge from different routes: altered initial fluctuation spectra, stochastic heating by external perturbers, or numerical relaxation. As a result, robust interpretation requires combining profile evolution with complementary diagnostics such as enclosed-mass evolution, shell-resolved kinematics, and sensitivity to solver controls.

This creates a central interpretive challenge for the present work: an apparent cusp-to-core transition can arise from physical mechanisms, from numerical artifacts, or from both acting simultaneously. The role of the theoretical background is therefore to define diagnostics that separate these possibilities as rigorously as possible.

2.3 Numerical heating and relaxation effects in N-body systems

An N-body simulation represents a continuous collisionless fluid with a finite number of massive macro-particles. This discretization introduces spurious collisionality: close encounters can produce artificial two-body relaxation, velocity diffusion, and long-term heating, especially in poorly resolved central regions (Binney and Tremaine, 2008; Power et al., 2003).

To first order, the local relaxation timescale scales approximately as

$$t_{\mathrm{relax}} \sim \frac{N}{8 \ln N}\, t_{\mathrm{cross}}, \tag{2.4}$$

where N is the number of effective particles supporting the region of interest and t_{cross} is a characteristic crossing time (Binney and Tremaine, 2008). This scaling emphasizes why central halo regions are vulnerable: they contain fewer particles and shorter dynamical times, making discreteness effects comparatively stronger.

The classical relaxation timescale argument provides a first-order test: if the numerical relaxation time at a given radius is not much longer than the simulated evolution time, then the resulting structure may be numerically biased. In practice, this means that central density trends must always be interpreted together with particle number, timestep control, force regularization, and force-accuracy parameters.

Power et al. translated these ideas into practical convergence guidance for halo simulations by linking reliability in the inner profile to a combination of mass resolution, timestep sampling, and force softening (Power et al., 2003). Their framework remains a benchmark because it moves beyond single-parameter tuning and treats convergence as a coupled numerical problem.

For this thesis, numerical heating is not treated as a generic warning but as an explicit hypothesis to test. If central density decline is dominated by relaxation artifacts, it should correlate with changes in numerical controls. If it persists across broad parameter changes, a more global dynamical mechanism becomes more plausible.

2.4 Gravitational softening: fixed and adaptive formulations

Gravitational softening modifies the short-range Newtonian force to prevent unphysical singular accelerations from particle-particle encounters. With a fixed softening length ϵ, one obtains a trade-off: large ϵ suppresses noise and relaxation but erases genuine small-scale structure; small ϵ improves nominal resolution but can increase discreteness-driven heating (Power et al., 2003; Zhang et al., 2019).

In many implementations, this can be written schematically with a softened pair force,

$$\mathbf{F}_{12} = -\frac{G m_1 m_2}{\left(r_{12}^2 + \epsilon^2\right)^{3/2}} \mathbf{r}_{12}, \tag{2.5}$$

which regularizes the r^{-2} singularity at very short separation (Zhang et al., 2019). The interpretation of ϵ is therefore two-fold: it is both a numerical stabilizer and a scale that limits physically meaningful force gradients.

Power et al. established influential convergence criteria linking softening, timestep, and particle number in the inner halo (Power et al., 2003). Later work showed that these prescriptions may be conservative for some setups, while also emphasizing that excessively small softening can bias the innermost density high (Zhang et al., 2019). Taken together, these studies show that softening is not a single "best" number independent of context.

Adaptive softening addresses this by varying effective force resolution with local conditions. However, naive implementations violate conservation laws. The Lagrangian formalism of Price and Monaghan demonstrated how to construct adaptive softening with the additional correction terms required for consistent energy and momentum conservation (Price and Monaghan, 2007). Modern conservative extensions generalize this framework to multi-species collisionless simulations (Hopkins et al., 2023).

From a methodological perspective, adaptive schemes are especially relevant for halos with strong radial density contrasts. A fixed ϵ that is adequate in the outskirts may be too coarse in the center, while an ϵ tuned to the center may be too noisy at large radii. Conservative adaptive formulations provide a way to mitigate this tension while preserving the invariants required for long integrations.

In the context of this thesis, softening is a key control variable, but not an isolated one. Its interpretation depends on the interaction between local force regularization and global gravitational coupling across the halo.

2.5 Gravity solver controls in SWIFT: Tree/FMM context

Beyond softening, force computation in modern cosmological codes depends on hierarchical gravity solvers (tree and fast multipole expansions) and their acceptance criteria. These methods accelerate long-range gravity by replacing exact pairwise summation with controlled approximations, whose errors depend on opening criteria and expansion tolerances.

This introduces a second numerical layer on top of softening. Even with identical initial conditions and identical softening, changes in opening criteria or multipole tolerances can alter force errors in a way that accumulates over many dynamical times. For slowly evolving central densities, small systematic differences can integrate into measurable secular trends.

In practical terms, parameters such as the multipole acceptance criterion (MAC), geometric opening thresholds (e.g. θ_{cr}), expansion tolerances (e.g. FMM error controls), and timestep factors set the balance between computational cost and force accuracy. Because central density evolution is sensitive to small cumulative force biases, these controls must be included in any robust sensitivity analysis of cusp/core evolution.

The same logic applies to timestep controls. If timesteps are too large compared to local orbital times, integration error can mimic diffusion or heating. If they are too small, the simulation becomes expensive but not necessarily more accurate unless force errors are also reduced. A reliable interpretation therefore requires coordinated tests across both integration and force-approximation parameters.

Therefore, when this thesis tests solver settings, the goal is not merely code tuning. It is a mechanism test: if the observed inner-density trend is primarily a solver artifact, changing these controls should alter either its onset or its long-term rate.

2.6 Why shell structure and mass redistribution matter

A central point for this thesis is that inner density evolution is not purely local. Even when interest focuses on the innermost region, the central potential is influenced by mass and orbital structure over a broad radial range. Particle populations at larger radii can contribute to central acceleration fluctuations, angular momentum exchange, and gradual redistribution of enclosed mass.

Recent theoretical work on stochastic tidal fields generated by clumpy substructure provides a useful interpretation for this non-local coupling. In these treatments, gravitational fluctuations induce diffusion in orbital integrals and can transfer energy and angular momentum through repeated perturbations rather than through single catastrophic events (Peñarrubia, 2019, 2017). This view is directly relevant to shell-based diagnostics: outer shells can influence central dynamics through cumulative, time-correlated forcing.

An important implication is that central density decline may be driven by the full halo environment, not only by the innermost force law. If true, changing local softening alone may leave a substantial part of the trend unchanged. This is precisely why shell-resolved measurements and cumulative mass diagnostics are central to the thesis design.

This motivates a shell-based analysis in which the halo is decomposed into radial regions and their dynamical influence on the center is tracked over time. If outer shells contribute substantially to central kinematic evolution, then changing only inner local parameters (for example, local softening) may have limited impact on the global trend.

This non-local perspective is consistent with previous findings in the project context, where persistent central density decline can remain visible across multiple numerical variations (Salamin, 2026). The theoretical implication is that diagnosis must combine local resolution tests with global mass-transport diagnostics.

2.7 Chapter summary

The theoretical framework developed here leads to three working principles for the rest of the thesis. First, cusp/core evolution in ultra-faint dwarfs is physically informative only when accompanied by explicit numerical convergence and solver-sensitivity checks. Second, gravitational softening is necessary but not sufficient to explain long-term inner-density evolution. Third, shell-to-shell coupling and cumulative mass redistribution provide a natural mechanism class that must be tested alongside local numerical effects.

Operationally, this means that the results chapter should be read as a sequence of discriminating tests. Each test changes one class of assumptions and asks whether the central trend persists, weakens, or disappears. A persistent trend across softening and solver variations strengthens the case for a global redistribution process; strong sensitivity to one parameter would instead indicate a dominant numerical origin.

The next chapter translates these principles into a concrete simulation and analysis framework, including parameter studies, centering strategy, density diagnostics, and shell-based measurements.

3 Simulation setup, framework and implementation

This chapter describes the numerical setup and analysis pipeline used to investigate the persistent central density decline identified in the previous chapters. The focus is on reproducibility: each subsection documents the simulation code configuration, initial-condition choices, diagnostic definitions, and implementation details used to test competing numerical and physical explanations.

The workflow combines (i) controlled collisionless N-body runs in SWIFT, (ii) systematic parameter variations of the self-gravity solver, (iii) profile and cumulative-mass diagnostics extracted from snapshots, and (iv) shell-based analyses designed to quantify non-local contributions to the dynamics of the innermost region.

3.1 SWIFT simulation setup

All simulations in this thesis are dark-matter-only and non-periodic, evolved with SWIFT in self-gravity mode (and external gravity when required by specific tests). In the working setup, runs are launched through

```
swift --self-gravity [--external-gravity] params.yml
```

with outputs consisting of snapshots, global statistics, and timestep logs.

The internal unit system is chosen so that

$$1 \text{ length unit} = 1 \text{ kpc}, \qquad 1 \text{ velocity unit} = 1 \text{ km s}^{-1}, \qquad 1 \text{ mass unit} \approx 10^{10}\, M_\odot, \tag{3.1}$$

which allows straightforward interpretation of profile radii (kpc), velocity dispersions (km/s), and enclosed masses in solar units after post-processing conversion.

3.1.1 Baseline gravity and time integration choices

The parameter-study baseline adopts the self-gravity configuration

$$\eta = 0.002, \qquad \text{MAC} = \text{geometric}, \qquad \theta_{\text{cr}} = 0.7, \qquad \epsilon_{\text{fmm}} = 10^{-3}, \tag{3.2}$$

with minimum and maximum timesteps typically constrained in the interval

$$\Delta t_{\min} = 10^{-6}, \qquad \Delta t_{\max} = 10^{-2} \tag{3.3}$$

To probe solver sensitivity, each of the following was varied while keeping the others fixed: timestep factor η, MAC choice (geometric/adaptive), geometric opening threshold θ_{cr}, and ϵ_{fmm}. These controlled tests form the foundation of the numerical robustness analysis presented in the results chapter.

3.1.2 Output cadence and bookkeeping

The production analysis relies on regular snapshot dumps that allow profile reconstruction at multiple times. In parallel, SWIFT writes global conservation and bulk quantities in `statistics.txt`, and timestep activity in `timesteps.txt`.

3.2 Halo initial conditions and parameter choices

The halo models are generated as collisionless particle realizations with controlled inner slopes to enable cusp/core comparisons under otherwise matched numerical conditions.

Initial conditions are produced with the pNbody code:

```
ic_gen_2_slopes --alpha 1 --beta 3 --Rmin 1e-3 --Rmax 50 --r_cutoff 50 \
    --rho0 0.0027658297046461644 --rs 0.828177997155062 \
    --ptype 1 --mass 61664.060524 -t swift -o halo.hdf5
```

which corresponds to a two-slope halo with tunable inner structure and an outer truncation radius.

Across the broader thesis campaign, the most relevant controlled parameters are:

- halo type: cusp and core initial conditions,
- box-size variants for finite-volume sensitivity checks,
- gravitational softening variants,
- self-gravity solver parameters (η, MAC, θ_{cr}, ϵ_{fmm}).

The practical philosophy is to keep one baseline run and modify one parameter family at a time, so each subsection in the results can be interpreted as a direct hypothesis test rather than a multi-parameter confounded comparison.

3.3 Density, shell, and mass diagnostics

This section defines the observables used throughout the thesis.

3.3.1 Spherically averaged density profiles

Density profiles are extracted from snapshots with a radial grid up to a user-defined $r_{\max}$ and N_r bins. In practice, the analysis scripts use both moderate and high radial resolutions, including $N_r = 500$ in the inner-region-focused runs.

For shell i with mass ΔM_i and boundaries $(r_{i,\text{in}}, r_{i,\text{out}})$, the estimator is

$$\rho_i = \frac{\Delta M_i}{\frac{4\pi}{3}\left(r_{i,\text{out}}^3 - r_{i,\text{in}}^3\right)}. \tag{3.4}$$

Profiles are then compared to the initial state using ratio curves $\rho(r,t)/\rho(r,t_0)$, which makes secular flattening or depletion immediately visible.

3.3.2 Softening-scale marker in profile plots

To separate physically interpretable radii from numerically delicate scales, profile plots include a vertical reference at

$$r_{\text{ref}} = 2.8\epsilon, \tag{3.5}$$

with ϵ the run softening length. This marker is not used as a hard cutoff, but as a visual guide when discussing central profile evolution.

3.3.3 Cumulative mass profiles

Cumulative mass diagnostics are computed via

$$M(< r) = \sum_{r_j<r} m_j, \tag{3.6}$$

including optional decomposition by particle type when needed. The implementation supports multiple centering modes (center of mass, median, manual center, and shrinking-sphere center), with shrinking-sphere used in the dedicated cumulative-mass tests because of its robustness to asymmetries and mild off-centering.

3.3.4 Velocity and anisotropy diagnostics

Using the same radial binning, the analysis computes velocity moments and anisotropy indicators, including

$$\beta(r) = 1 - \frac{\sigma_t^2}{2\sigma_r^2}, \tag{3.7}$$

as complementary diagnostics for identifying orbital heating and redistribution mechanisms.

3.4 Post-processing pipeline and centering

The post-processing pipeline is designed to keep the density measurements stable against two common sources of noise: snapshot-to-snapshot variance and small centering offsets. These steps are essential for ensuring that the measured trends reflect the simulation rather than the extraction procedure.

3.4.1 Snapshot concatenation

To reduce stochastic noise in the density-evolution curves, consecutive snapshots are concatenated into larger time blocks before the profile comparison is made. This averaging procedure smooths out short-timescale fluctuations that appear in individual outputs, while preserving the secular evolution of the halo.

The key point is that concatenation changes the visual smoothness of the curves, but not the underlying behaviour: the declining inner-density trend remains present after the time blocks are combined. For that reason, concatenation is used only as a noise-reduction step, not as a way of altering the physical interpretation.

3.4.2 Shrinking-sphere centering

Centering is handled independently from the density estimation. Each snapshot is recentered using a shrinking-sphere algorithm, which iteratively refines the halo center by repeatedly restricting the particle set to smaller radii around the current estimate. This method is robust against mild asymmetries and against the gradual drift that can occur in long collisionless runs.

Using the shrinking-sphere center ensures that the innermost bins of the density profile are measured relative to a dynamically meaningful origin. In practice, this removes one of the most obvious numerical artefacts that could otherwise be mistaken for a central density decline.

3.5 Shell-based analysis framework

To determine whether the central density decline is driven purely by local inner-halo physics or by non-local coupling from the broader halo structure, we decompose the halo into radial shells and quantify each shell's contribution to the gravitational potential and dynamics of the innermost region.

3.5.1 Motivation and design principle

In a spherically symmetric system, the radial force on a particle at radius r depends only on the enclosed mass $M(<r)$ by the shell theorem. However, the gravitational potential depends on all mass, including material at larger radii. This distinction is crucial: outer shells do not change the local force gradient, but they do deepen the potential well and contribute to the escape-speed scale. The shell-based analysis tests whether the central decline is sensitive to the cumulative influence of outer material on the inner potential structure.

The analysis addresses a specific question: if outer shells contribute substantially to the inner potential depth, then a hybrid model that replaces the outer halo ($r > r_{\rm cut}$) with an analytic potential while retaining the inner region ($r < r_{\rm cut}$) as particles should reproduce similar evolutionary trends. Conversely, if the central decline persists even when the outer dynamics is replaced by a smooth analytic model, then the effect is either driven by inner-region physics or by the potential structure rather than by dynamical interactions with individual outer particles.

3.5.2 NFW truncated-halo model

The analysis uses a spherically symmetric Navarro–Frenk–White (NFW) halo model (Navarro et al., 1996), which is a standard benchmark for dark matter halos:

$$\rho(r) = \frac{\rho_s}{(r/r_s)(1+r/r_s)^2}, \tag{3.8}$$

where ρ_s is the characteristic density and r_s is the scale radius. The enclosed mass within radius r is:

$$M(<r) = 4\pi\rho_s r_s^3 \left[\ln(1+x) - \frac{x}{1+x}\right], \quad x = \frac{r}{r_s}. \tag{3.9}$$

The halo is truncated at a virial radius $r_{\rm vir}$. The gravitational potential including the truncation is:

$$\Phi(r) = -\frac{GM(<r)}{r} - 4\pi G\rho_s r_s^2 \left[\frac{1}{1+r/r_s} - \frac{1}{1+r_{\rm vir}/r_s}\right], \tag{3.10}$$

and the radial acceleration (force per unit mass) is:

$$a(r) = \frac{GM(<r)}{r^2}. \tag{3.11}$$

3.5.3 Key diagnostics and definitions

The analysis is built around four central diagnostic quantities, each addressing a different aspect of halo structure:

Enclosed mass and shell mass fraction The enclosed mass $M(<r)$ tracks how the total mass budget grows with radius. The shell mass fraction is the mass in a radial shell divided by the total halo mass, showing the radial distribution of the mass budget. For an NFW profile, the enclosed mass grows steeply near the center but the shell mass fraction increases toward larger radii, indicating that the outer halo contains a substantial fraction of the total mass.

Central potential depth from individual shells For each shell at radius range $[r_i, r_{i+1}]$, we calculate its contribution to the gravitational potential at the center ($r = 0$). This is given exactly by:

$$\Delta\Phi_{\rm shell}(0) = -4\pi G\rho_s r_s^2 \left[\frac{1}{1+r_i/r_s} - \frac{1}{1+r_{i+1}/r_s}\right]. \tag{3.12}$$

Shells well outside the center still make sizeable contributions to the potential depth at $r = 0$ (and therefore to the escape-speed scale $v_{\rm esc} \propto \sqrt{-2\Phi}$). The fractional contribution of each shell quantifies this effect.

Outer-potential fraction For a given probe radius $r_{\rm probe}$ and cutoff radius $r_{\rm cut}$, the outer-potential fraction is defined as:

$$f_{\rm outer}(r_{\rm probe}, r_{\rm cut}) = \frac{\left|\Phi_{>r_{\rm cut}}(r_{\rm probe})\right|}{\left|\Phi_{\rm total}(r_{\rm probe})\right|}, \tag{3.13}$$

where $\Phi_{>r_{\rm cut}}$ is the potential contribution from material outside the cutoff and $\Phi_{\rm total}$ is the total potential at that radius. This measures how strongly the outer halo influences the local potential at the probe radius. By varying $r_{\rm cut}$, we can determine the radius beyond which the outer halo becomes a small correction.

Force contribution by shell The radial force (acceleration) at a probe radius $r_{\rm probe}$ is:

$$a(r_{\mathrm{probe}}) = \frac{GM(< r_{\mathrm{probe}})}{r_{\mathrm{probe}}^2}. \tag{3.14}$$

Under spherical symmetry, only mass interior to r_{probe} contributes. This diagnostic confirms that outer shells, despite deepening the potential, do not directly modify the local force field—a consistency check on the shell-theorem interpretation.

3.5.4 Implementation: analytical NFW shell decomposition

Rather than relying on a numerical computation of the integrals, the shell analysis uses exact analytic formulas for a truncated NFW profile. The halo is decomposed into logarithmically spaced spherical shells:

$$r_i = r_{\mathrm{min}} \left(\frac{r_{\mathrm{vir}}}{r_{\mathrm{min}}} \right)^{i/N_{\mathrm{shells}}}, \quad i = 0, 1, \ldots, N_{\mathrm{shells}}, \tag{3.15}$$

where typically $N_{\mathrm{shells}} = 80$ and $r_{\mathrm{min}} \approx 0.02$ kpc.

For each shell $[r_i, r_{i+1}]$, exact closed-form expressions compute: (1) shell mass $M_{\mathrm{shell}} = M(< r_{i+1}) - M(< r_i)$, (2) shell contribution to potential at any radius, and (3) cumulative potential contributions for integration over shells.

The analytical approach avoids numerical-integration noise and ensures all diagnostics are computed to high precision. The Python implementation uses these formulas to generate diagnostic plots showing: (i) enclosed mass and shell mass-fraction distribution, (ii) shell contributions to the central potential depth, (iii) outer-potential fraction as a function of cutoff radius, and (iv) a 2D map of outer-fraction versus both probe and cutoff radii.

3.6 Numerical implementation of the hybrid model

Motivated by the shell results, a dedicated hybrid gravity model was implemented and tested on a custom SWIFT branch (`truncated-NFW`). The model enforces a hard radial split at a cutoff radius R_{cut}:

- inner region ($r \leq R_{\mathrm{cut}}$): standard self-gravity, no external NFW potential,
- outer region ($r > R_{\mathrm{cut}}$): external NFW potential applied, with optional masking of dark-matter self-gravity sourcing outside the cutoff.

The design goal is not to claim a final physical model, but to construct a targeted numerical experiment: if outer-halo coupling is a major driver of central evolution, changing how the outer region sources gravity should produce a measurable response in inner-density trends.

3.6.1 Implementation objectives and design choices

The truncated-NFW backend is designed to isolate the dynamical role of the outer halo while keeping the inner region fully self-consistent. The implementation follows three guiding principles:

- **Clear separation of responsibilities**: inner particles ($r \le R_{\rm cut}$) evolve under the usual SWIFT self-gravity solver and provide the only resolved self-gravity sources inside the cutoff. Outer particles ($r > R_{\rm cut}$) evolve in the analytic truncated-NFW field, with resolved gravity contributions removed by construction.
- **Minimal invasive changes**: the feature is implemented as an optional, compile-time selectable external-potential backend and does not alter the default gravity code paths when disabled.
- **Deterministic, documented boundary convention**: the hard boundary is defined as inner for $r \le R_{\rm cut}$ and outer for $r > R_{\rm cut}$, avoiding ambiguous behaviour at the cutoff.

3.6.2 Build-time wiring and backend selection

The backend is registered as an external-potential option and selected at configure time. The recommended configure invocation is:

```
./configure --with-ext-potential=truncated-nfw
```

At build-time the configure script defines the macro identifying the truncated-NFW backend and compiles the corresponding header implementation. This approach keeps the extra code path isolated behind a single configure flag and reduces the risk of accidental behavior changes in other backends.

Files involved at build time include the project's `configure.ac` (backend option and macro definition), `src/potential.h` (backend include branch), and `src/potential/truncated_nfw/potential.h` (backend implementation). Gravity-side hooks live in `src/gravity/*`, `src/multipole.h`, and mesh-gravity files, but remain dormant unless the truncated-NFW backend is selected.

3.6.3 Runtime parameters and YAML configuration

Runtime control is provided through the existing `NFWPotential` block in the parameter YAML. The truncated-NFW mode extends the canonical NFW keys with a small set of required and optional keys:

- `cutoff_radius`: required floating-point value; the radial cutoff $R_{\rm cut}$ in kpc.

- `dm_mask_outside_cutoff`: optional boolean (default 1). If true, DM outside the cutoff is excluded as a source of self-gravity.
- `position`: center of the external potential (array); default behaviour follows the usual `NFWPotential:useabspos` semantics.
- `useabspos`: boolean as in the standard NFWPotential, controls interpretation of `position`.

The parameter parser in the backend validates that `cutoff_radius` is present when truncated-NFW is selected and prints the effective setup at startup (center, halo parameters, cutoff, masking flag). The same center definition is propagated to the masking logic through dedicated globals (mask enabled flag, mask center, and R^2_{cut}), so potential and masking use exactly the same geometric reference.

3.6.4 Two-layer masking logic: sources and targets

The gravity split is enforced through two complementary mechanisms.

Source-side masking. Dark-matter particles outside the cutoff use an effective gravitational source mass of zero through `gravity_get_effective_mass_from_position()`. Operationally this corresponds to `grav_mass_factor=0` for outer DM and `grav_mass_factor=1` otherwise. The effective mass is then consumed consistently by pairwise interactions, gravity caches, multipole construction, and no-cache pair loops (including `runner_doiact_grav.c`), so outer DM does not source resolved self-gravity.

Target-side suppression. Outer particles are also prevented from receiving resolved self-gravity updates. This ensures that the outer domain is analytic-only not only as a source prescription but also at the level of target acceleration/potential write-back.

To implement this, a common geometric helper `gravity_is_outside_truncated_nfw_cutoff(const double x[3])` was added in both gravity variants: `src/gravity/Default/gravity.h` and `src/gravity/MultiSoftening/gravity.h`.

This helper is then used to gate resolved-gravity write-back paths:

- `src/gravity_cache.h`: in `gravity_cache_write_back(...)` a guard skips adding cached pair/tree acceleration and potential to outer targets;
- `src/multipole.h`: in `gravity_L2P(...)` an early return prevents multipole target updates for outer particles;
- `src/mesh_gravity.c`: in `mesh_to_gpart_CIC(...)` an early return suppresses local PM interpolation for outer particles;
- `src/mesh_gravity_mpi.c`: in `mesh_patch_to_gparts_CIC(...)` an early return suppresses MPI PM interpolation for outer particles.

Operationally, for any target with $r > R_{\mathrm{cut}}$, resolved pair/tree/mesh contributions are not written to the particle state. Outer particles therefore evolve only under analytic external potential terms.

The same predicate is used in both local and MPI-distributed mesh paths, avoiding rank-dependent behaviour at the cutoff surface.

3.6.5 External potential behaviour and timestep coupling

The external-potential API is responsible for returning three quantities for each particle position: acceleration $\mathbf{a}_{\mathrm{ext}}(\mathbf{x})$, potential energy contribution $\Phi_{\mathrm{ext}}(\mathbf{x})$, and optional timestep constraint multipliers. In truncated-NFW mode these are implemented as

- for $r \le R_{\mathrm{cut}}$: $\mathbf{a}_{\mathrm{ext}} = 0$, $\Phi_{\mathrm{ext}} = 0$ (no external field acting on inner particles),
- for $r > R_{\mathrm{cut}}$: standard truncated NFW analytic expressions for $\mathbf{a}_{\mathrm{ext}}$ and Φ_{ext} are returned.

The analytic potential/acceleration are injected through the standard external-potential hooks (including `gravity_add_comoving_potential(...)`), while resolved-gravity suppression is handled in gravity and mesh write-back paths.

This separation clarifies potential bookkeeping: outer particles no longer accumulate resolved self-potential from tree/pair/mesh, but they still receive the analytic external contribution via the external backend and enter $E_{\mathrm{pot,ext}}$ diagnostics through `external_gravity_get_potential_energy(...)`.

The external potential may optionally supply a timestep multiplier for outer particles (for example to limit timesteps when the analytic field varies strongly), but by default the domain's existing timestep criteria govern integration.

3.6.6 Centering, cutoff convention and edge cases

The backend uses a single, explicitly provided center for both the NFW potential and the cutoff test. Two configuration choices affect this center: absolute coordinates (`useabspos=1`) or coordinates relative to box origin/periodic convention (`useabspos=0`). The policy is documented at startup and logged.

We adopt the hard-step convention: particles at exactly $r = R_{\mathrm{cut}}$ are classified as inner ($r \le R_{\mathrm{cut}}$). This avoids oscillatory reclassification for particles grazing the boundary and is consistent with the mathematical definitions used in the analytic diagnostics.

3.6.7 Serialization, restart safety and diagnostics

All runtime parameters are recorded in the standard external-potential state so that restarts reproduce the mask and potential choices. On startup the backend prints a compact diagnostics summary that includes backend name, chosen center, r_{cut}, and whether DM masking is active. The initialization path that sets masking state and effective masses is executed at fresh start and restart, and is refreshed at the gravity update cadence (per step or per tree/domain rebuild, depending on run configuration).

Because the modification changes the effective source mass (and therefore total self-gravity source) the global energy bookkeeping routines are instrumented to report the expected non-conservative terms and to show the separation between particle-sourced and analytic-potential contributions to the total potential energy. These diagnostics are intended for validation and are not enabled in tight production runs by default.

3.6.8 Foreign-particle and MPI consistency

To preserve exact force symmetry across MPI boundaries, the effective mass semantics must be identical for both local and foreign particles. The implementation therefore applies one of two consistent strategies when packing remote particles:

1. export the effective mass (m_{eff}) directly in the packed structure, or
2. export both the raw mass and the mask-factor, and reconstruct m_{eff} on the receiving side.

Either strategy yields the same physics if used consistently; the implementation enforces a single semantics for local and remote particles so multipole and pairwise forces remain symmetric across rank boundaries.

3.6.9 Validation strategy

The implementation is accompanied by a small validation suite that exercises the key correctness properties:

- single-particle checks: place test particles just inside and just outside R_{cut} and verify that inner particles experience no external acceleration while outer particles feel the analytic NFW acceleration;
- self-gravity exclusion: confirm that masked outer particles do not contribute to monopole moments in the tree and that pairwise P2P forces from outer sources are absent (beyond numerical tolerance);

- MPI consistency: run identical initial conditions on 1 and on multiple ranks and verify that accelerations and early-time trajectories near the cutoff are numerically identical within roundoff;
- restart repeatability: run to a snapshot, restart, and confirm the mask and potential behaviour are unchanged.

These checks are documented and included in the project's validation notes so the truncated-NFW backend can be accepted for experimental use only after passing the suite.

3.6.10 Caveats and limitations

The present implementation aims at a minimal, well-defined numerical experiment and therefore imposes several deliberate limitations:

- the cutoff is a hard step; no smooth transition shell is implemented (this can be added later if numerical artefacts at the boundary are observed),
- only dark-matter particles are eligible for masking; gas, stars and sink/BH particles keep their full source mass, and their coupling to the external potential is unchanged,
- using a fixed centre for the external potential while the particle distribution drifts can create a mismatch; the user must therefore choose the centre carefully (e.g. by using a shrinking-sphere centre estimate before launching a hybrid run),
- energy is not strictly conserved across the particle/analytic split because masked particle mass is removed from the self-gravity source and replaced by an analytic potential; diagnostics are provided to quantify this effect,
- for strict energy continuity diagnostics, a constant potential offset at R_{cut} may still be desirable in the external backend (forces are unaffected by this constant, but reported potential energies can shift).

Implementation details and exact file locations are documented in the project plan and inline in the code (see `software/swiftsim/src/potential/truncated_nfw/potential.h` and related gravity hooks). The repository includes the configure option, the backend header, and the gravity-kernel adaptations required to run and validate the hybrid experiments described in Chapter 3.5.

4 Results and discussion

This chapter presents the numerical results obtained from the reference simulation campaign and the subsequent sensitivity studies. The central question is whether the persistent decline of the inner density is an intrinsic dynamical feature of the halo, a consequence of numerical settings, or an artefact of the analysis procedure. The discussion is therefore structured around a sequence of increasingly restrictive checks: first the reference evolution itself, then the post-processing pipeline, then controlled variations of the simulation volume, the initial halo structure, the softening length, and the gravity solver settings.

The completed sections below are written as a scientific discussion rather than as a notebook-style log. The later sections remain intentionally unfinished and are marked with explicit placeholders.

4.1 Baseline behaviour

The baseline run shows a clear cusp-to-core transition, but the core does not stabilize. Instead, the inner density continues to decline throughout the simulated time span. This is the central empirical result of the thesis. In the broader context of halo evolution, the literature shows that collisionless systems can display a wide range of inner responses to external perturbations and time-dependent forcing, including stochastic heating by substructure and repeated diffusion in energy–angular-momentum space (Peñarrubia, 2017, 2019). Recent controlled N-body models by Errani et al. show that collisional star–dark-matter energy exchange can heat a cusp into a core and that this evolution tends to self-regulate as central dark-matter densities drop and binary formation increases, i.e. the contraction/heating rates slow at late times (Errani et al., 2026). If that late-time stabilization picture applied directly to our setup, the inner density would be expected to flatten after core formation rather than continue declining. The reference run is therefore notable not because it forms a core, but because the core remains dynamically active rather than approaching a steady inner state.

The result is also instructive from the point of view of numerical convergence. Power et al.

showed that the inner halo is especially sensitive to the interplay between resolution, force softening, and timestep control (Power et al., 2003), while Zhang et al. later demonstrated that too aggressive a reduction of the softening length can artificially raise the innermost density (Zhang et al., 2019). The reference simulation does not display a simple monotonic dependence on any one of these parameters, which already suggests that the behaviour is not a trivial one-parameter resolution effect.

The baseline density-evolution plot and the snapshot sequence show the same qualitative pattern: the inner profile becomes shallower, but the inner density does not settle. The new gradient plot in Figure 4.1 makes this especially clear: the central density drops rapidly at early times, then continues to drift downward more slowly, without ever forming a stable plateau. The companion plot in Figure 4.2 presents the same evolution over a shorter time interval and adds the ratio to the initial conditions, which makes the secular decline easier to read at a glance. In that sense, the evolution resembles a continuing redistribution of mass rather than a completed structural transformation. This is the key point that motivates the rest of the chapter.

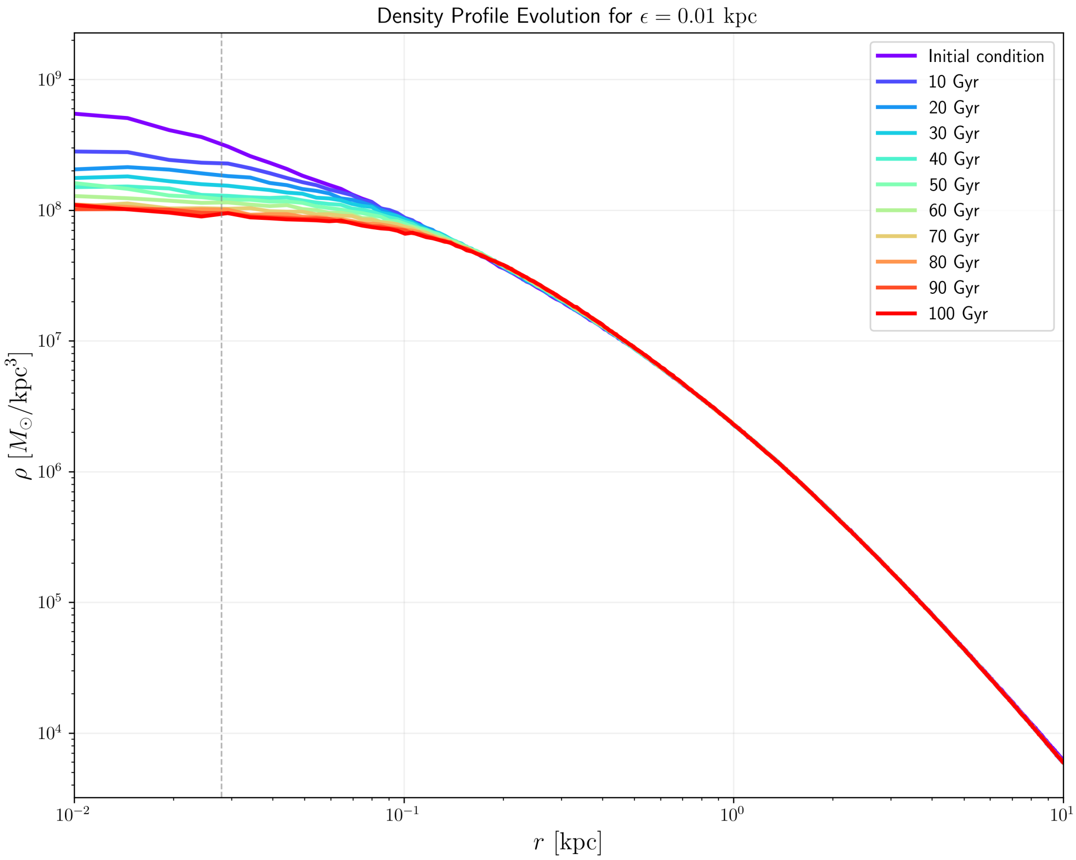

Figure 4.1: **Baseline density-profile evolution.** The central density decreases steadily over time, with the steepest change occurring early and the later evolution approaching a shallower, but still declining, trend. The gradient structure is consistent with a secular redistribution rather than a settled core.

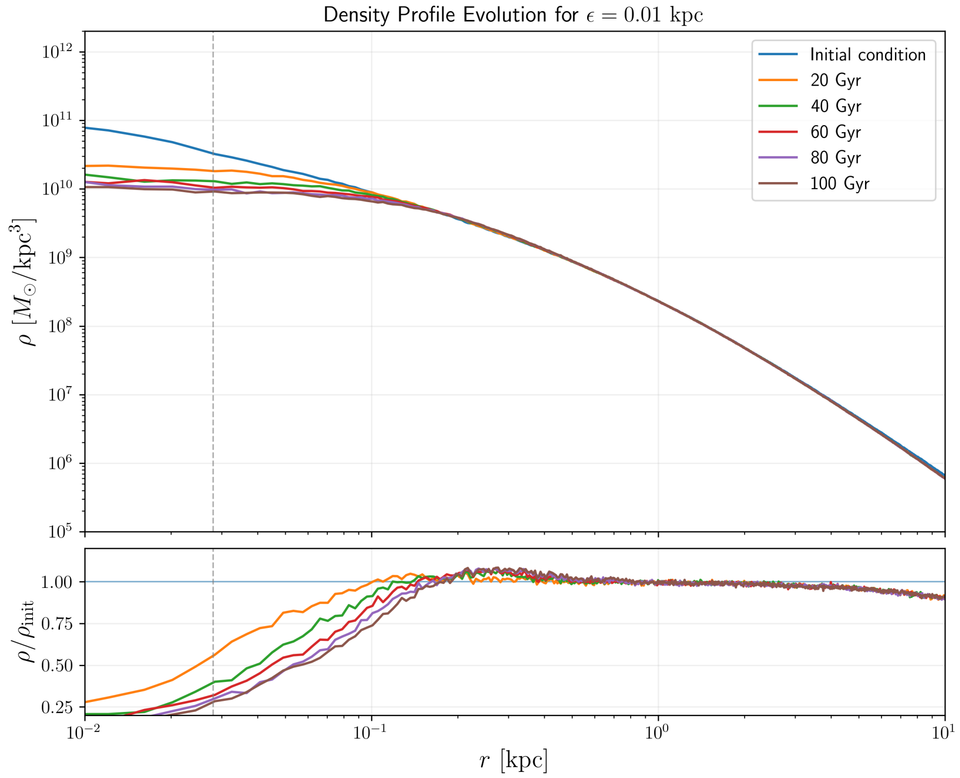

Figure 4.2: **Baseline evolution with ratio to the initial state.** The shorter time window clarifies the early-time evolution, and the lower panel shows the density ratio relative to the initial conditions. This representation highlights that the inner decline is continuous rather than a brief transient.

The comparison with the literature is therefore important: the simulation reproduces a cusp-to-core transition, but it does not reproduce a long-lived, stationary core. Whatever mechanism is responsible must therefore operate over the full duration of the run.

The most conservative conclusion at this stage is that the halo does not relax into a static inner structure on the simulated timescale. The observed behaviour is a secular evolution of the center, not merely an initial transient.

4.2 Box size

The role of the numerical domain was tested explicitly by repeating the reference experiment in three increasingly larger simulation volumes while holding all physical and numerical parameters fixed (identical particle realization, softening, timestepping rules and gravity solver settings). The objective was to determine whether finite-domain effects, long-range tidal contributions or boundary-condition artefacts can explain the persistent central-density decline.

Figure 4.3 summarises the outcome. The central-density diagnostic used throughout this thesis (the inner radial bin of the standard profile grid; see Section 3.3) shows the same secular decline in each volume. Enlarging the domain alters the quantitative rate of depletion by a modest amount, but it does not halt or reverse the process: the qualitative behaviour is invariant to the choice of box size.

There are two simple physical interpretations of this result. First, long-range mass beyond the immediate inner region changes the background potential and can therefore modify rates (consistent with the shell-decomposition analysis and the shell-theorem distinction between force and potential; see Chapter 3.5 and Binney and Tremaine 2008). Second, if the mechanism were purely an artefact of a truncated domain (for example a spurious boundary-driven tide or reflection), one would expect the effect to vanish as the box is enlarged; the absence of such a disappearance argues against a boundary-driven origin. These conclusions are in line with standard convergence diagnostics for collisionless codes (Power et al., 2003) and with the expectation that stochastic long-range perturbations modulate rates rather than create qualitatively new inner dynamics (Peñarrubia, 2017, 2019).

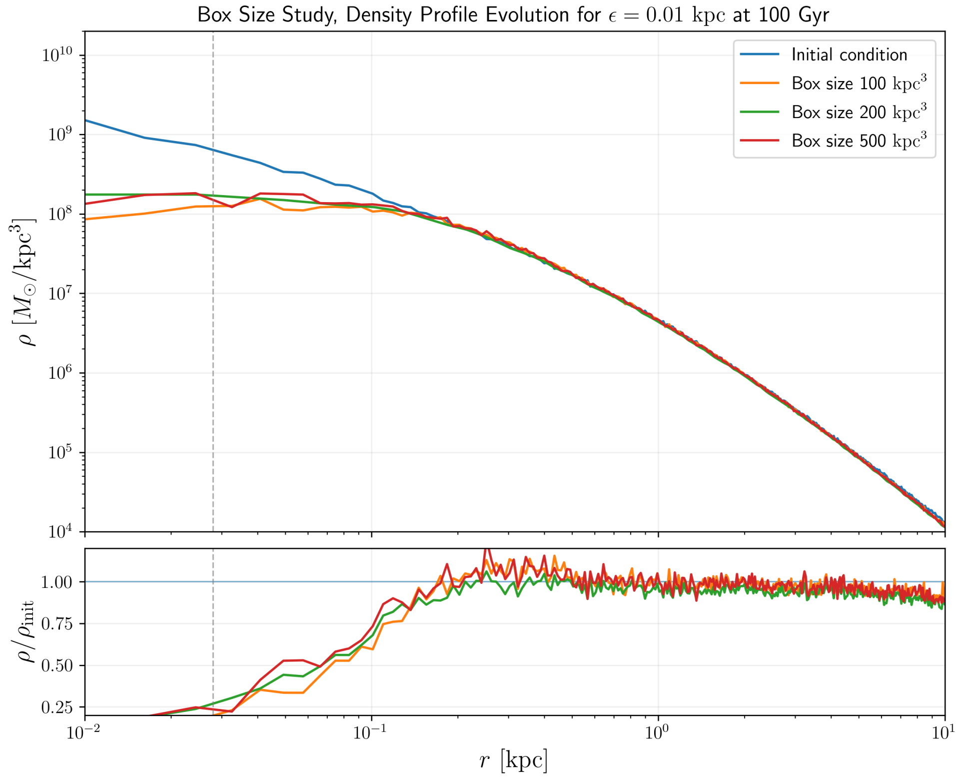

Figure 4.3: **Box-size study.** Comparison of the three tested simulation volumes (small, intermediate, large). The central-density diagnostic (inner radial bin) shows the same secular decline across volumes; changing the box size modifies the depletion rate only modestly.

In summary, the box-size experiment removes a simple numerical-domain explanation for the observed central decline: while the environment (volume size and outer mass distribution) can regulate the timescale, it is not the primary cause of the effect observed in the reference runs.

4.3 Initial halo structure

The comparison between cuspy and cored initial conditions is particularly informative because it distinguishes between a cusp-specific instability and a more general process. The core run still exhibits a decline of the central density, but the amplitude is smaller than in the cuspy case. This is a strong result because it means the mechanism is not limited to the special case of an initially steep central profile.

In physical terms, this suggests that the system is sensitive to the presence of a central potential well, but not exclusively to its initial logarithmic slope. A cored halo has less extreme inner structure and therefore reacts less strongly, yet the evolution remains qualitatively the same. This is particularly important in light of the Errani et al. result that cusp-heating evolution should weaken once the core is established and central dark-matter densities are reduced (Errani et al., 2026). In that scenario, a pre-existing core should not continue to lose central density over long timescales; however, in our runs it does. That mismatch reinforces the conclusion that the persistent decline cannot be explained as a simple cusp-to-core transient and is instead tied to a broader dynamical mechanism.

The literature on substructure-driven heating is again relevant here. Stochastic forcing does not require a cusp; it requires a fluctuating potential and a population of perturbers that can exchange energy with the central region (Peñarrubia, 2017). The persistence of the effect in the core run is therefore compatible with a non-local heating picture and inconsistent with the idea that the problem is merely a cusp pathology.

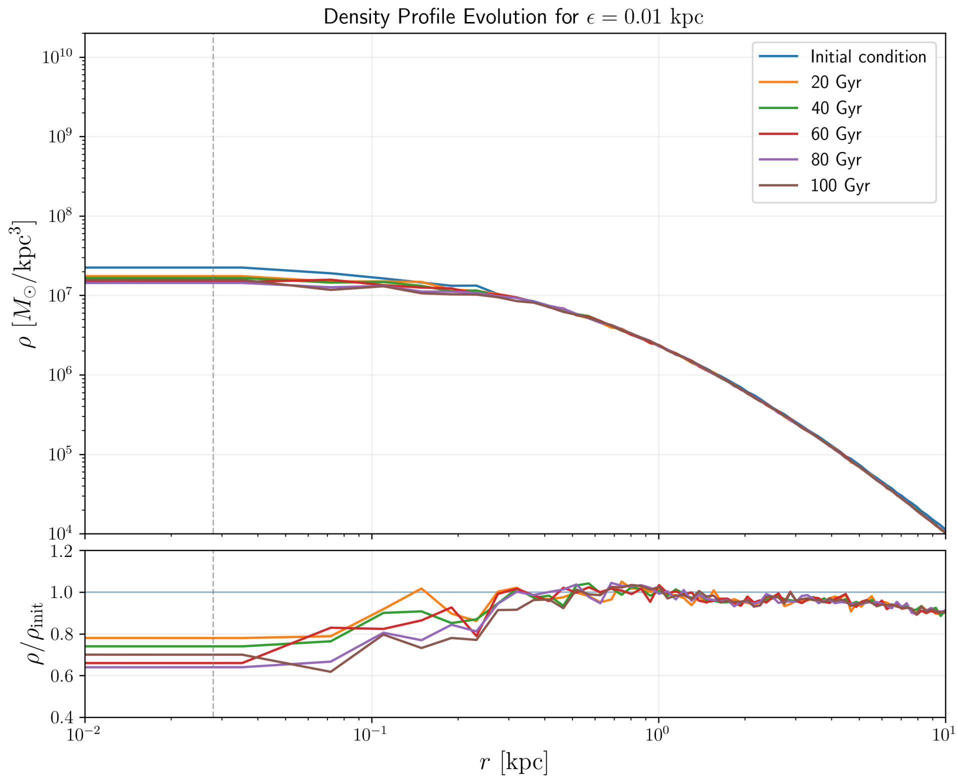

Figure 4.4: **Core initial-condition test.** The cored halo also shows a decline in central density, although with a smaller amplitude than the cuspy case. This directly contradicts a simple expectation of core formation followed by stabilization.

The practical implication is that changing the inner slope reduces the amplitude but not the mechanism. The central decline is therefore not restricted to cuspy initial conditions.

4.4 Gravitational softening

Softening is the most obvious numerical parameter to suspect in any study of the inner halo, because it directly regularizes short-range interactions. The literature makes clear, however, that the role of softening is subtle: too much softening suppresses genuine small-scale structure, while too little can artificially increase the density in the innermost region (Power et al., 2003; Zhang et al., 2019). In the convergence analysis of Zhang et al., the optimal softening length is the one that balances force bias against discreteness noise; below that scale, the central profile can become biased high because close encounters are less regularized and the inner region responds more strongly to numerical heating rather than to a genuine improvement in resolution. The present results fit that picture in the sense that the softening length changes the detailed rate of evolution, but not the existence of the decline itself.

This is an important negative result. If softening were the primary cause, the central density evolution should have shown a qualitatively different response when the softening length was varied. Instead, the trend persists across the tested range. The smallest tested softening is especially informative: it shows a modest increase in density as one approaches the center, but that should not be interpreted as a real improvement of the physical problem. The more likely explanation is that the reduced softening has allowed stronger numerical heating and a slightly more concentrated inner profile, which is exactly the kind of bias that an under-softened calculation can introduce (Power et al., 2003; Zhang et al., 2019).

The notes from the shell analysis already point in the same direction: outer shells contribute significantly to the central dynamics, and changing the local softening length does not alter the fact that those shells are still present and still exert gravity. In other words, softening can change the numerical regularization of pairwise forces, but it does not remove the mass distribution that sources the global potential.

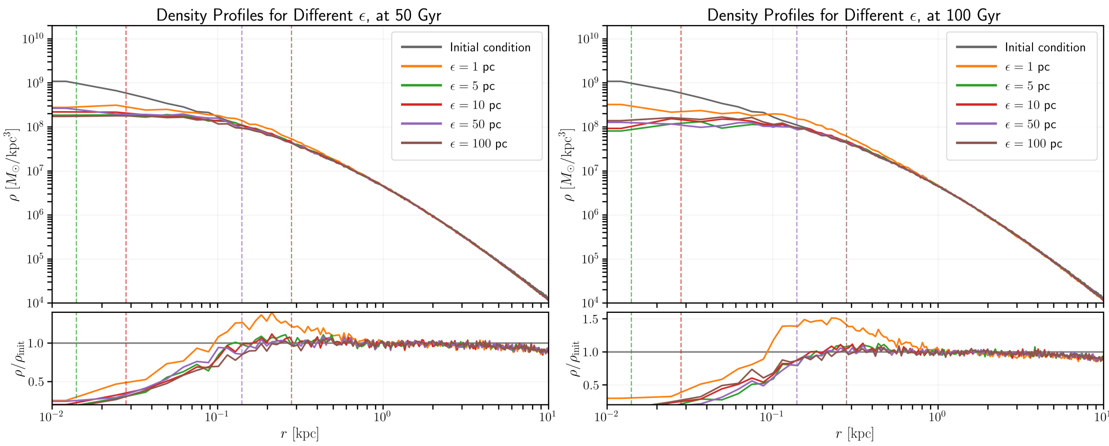

Figure 4.5: **Softening study.** Comparison of the same set of tested softening lengths at 50 Gyr (left) and 100 Gyr (right). The shorter-time view highlights the earlier evolution, while the longer-time view shows that the same hierarchy of softening runs follows the same qualitative trend. The smallest softening still shows a slight central-density increase close to the center, which is better interpreted as numerical heating than as a real improvement of the physical problem.

The conclusion is that softening is not the dominant driver. It is part of the numerical control budget, but not the mechanism behind the secular central depletion.

4.5 Self-gravity settings

The solver-parameter study addresses a more subtle question: whether the observed decline could be the cumulative effect of force-approximation or integration errors. This is a natural concern in hierarchical gravity solvers, where the acceptance criterion and multipole tolerances determine how accurately long-range interactions are represented. In principle, a slightly looser or tighter force approximation can shift the secular evolution of a halo even when the initial conditions are identical.

The present study varies four key control parameters of the self-gravity solver:

- **Timestep factor** η: Controls the integration timestep via $\Delta t = \eta \times t_{\rm dyn}$. A smaller η means finer temporal resolution and more frequent force updates; a larger η allows fewer timesteps and can accumulate more integration error over each step.
- **Multipole Acceptance Criterion (MAC)**: Determines when the force from a distant cell can be approximated by its multipole moments rather than evaluated exactly from individual particles. The *geometric* MAC uses the cell size and distance; the *adaptive* MAC adjusts the criterion based on the local particle distribution. Geometric MAC is faster but less accurate for clumpy structures.

- **Geometric opening angle** $\theta_{\rm cr}$: Specifies the threshold for geometric MAC: cells with $\theta_{\rm cr} = r_{\rm cell}/d < \theta$ activate multipole approximation. Smaller $\theta_{\rm cr}$ means a stricter criterion, forcing more exact evaluations and improving accuracy at higher computational cost.
- **FMM tolerance** $\epsilon_{\rm fmm}$: Controls the precision of the Fast Multipole Method used to compute far-field forces. Smaller tolerances mean higher precision, but smaller $\epsilon_{\rm fmm}$ requires more multipole terms and increases computation time.

The parameter range explored here is broad enough to reveal systematic sensitivity if it exists. The timestep factor was varied by roughly an order of magnitude, the MAC was changed between geometric and adaptive formulations, $\theta_{\rm cr}$ was tightened in the geometric case, and $\epsilon_{\rm fmm}$ was varied over two to three orders of magnitude. In a purely numerical explanation, one would expect at least one of these settings to have a decisive effect on the inner profile.

4.5.1 Timestep study

The timestep factor η determines the frequency of gravity and kinematics updates. Increasing η allows larger timesteps and reduces the number of force evaluations per unit time; decreasing η enforces finer temporal sampling and can improve phase-space accuracy. The baseline value of $\eta = 0.002$ is typical for high-resolution dark-matter studies, but it is instructive to test both more conservative ($\eta = 0.0002$, roughly $10\times$ finer) and more permissive ($\eta = 0.02$, roughly $10\times$ coarser) variants.

If timestep error were the main driver of central decline, the coarser run should exhibit notably faster or more severe depletion, while the finer run might show stabilization. That is not observed.

4.5.2 Multipole Acceptance Criterion study

The MAC choice reflects a fundamental trade-off in hierarchical tree methods: speed versus local accuracy. The geometric MAC is computationally cheaper but treats all distant cells identically; the adaptive MAC is more expensive but adapts the multipole threshold based on local structure. For a collisionless halo, the choice of MAC affects the accuracy of long-range forces and can subtly alter how external shells couple to the center.

If MAC were the primary sensitivity, changing between geometric and adaptive should produce a qualitatively different central-decline rate. The runs tested here show no such distinction.

4.5.3 Geometric opening angle study

For the geometric MAC, a smaller opening angle $\theta_{\rm cr}$ means stricter multipole approximation: more cell pairs are forced into exact evaluation rather than multipole summation. The baseline value $\theta_{\rm cr} = 0.7$ is a standard choice; the test value $\theta_{\rm cr} = 0.5$ represents a more stringent criterion and therefore higher force accuracy.

Lowering θ_{cr} should increase computational cost but also increase force precision. If force-approximation error were limiting the simulation, tightening θ_{cr} should reduce or delay the central decline. The observed effect is marginal.

4.5.4 FMM tolerance study

The Fast Multipole Method tolerance ϵ_{fmm} is a global precision parameter for the multipole expansion. The baseline value is $\epsilon_{fmm} = 0.001$ (relatively tight), and the test suite includes both looser values (0.01 and 0.1, representing 10× and 100× relaxation) and a tighter value (0.0001, representing 10× tightening). This range is designed to unambiguously reveal whether force approximation is a controlling factor.

Some runs with modified ϵ_{fmm} exhibit slightly improved early-time behaviour, particularly the tighter tolerance, but even those runs exhibit long-term central decline. The change is not enough to alter the qualitative long-term trend.

4.5.5 Summary table of self-gravity parameter variations

Table 4.1: Summary of self-gravity solver parameter values in the study. Each row lists the parameter, its baseline value (used in the reference run), and the test values explored to assess sensitivity.

Parameter	Baseline	Test values		
η (timestep)	0.002	0.0002	0.02	—
MAC	geometric	adaptive	—	—
θ_{cr} (geometric only)	0.7	0.6	0.5	—
ϵ_{fmm} (FMM tolerance)	0.001	0.0001	0.01	0.1

4.5.6 Overall interpretation of solver sensitivity

That is not what we observe across the full parameter set. The central density evolution remains broadly similar across all tested settings. This is consistent with the general lesson from convergence studies: changing a gravity control parameter may reduce error bars or shift the onset of bias, but it does not necessarily eliminate the physical or numerical process that dominates the evolution (Power et al., 2003; Zhang et al., 2019).

The most useful interpretation is therefore not that the solver settings are irrelevant, but that they are not the principal explanation. Their influence is secondary and limited to rate-level effects rather than qualitative behaviour. A solver parameter would need to have produced a qualitatively different central decline — either eliminating it or dramatically accelerating it — to be considered a dominant factor. None of the parameters tested here achieve that threshold.

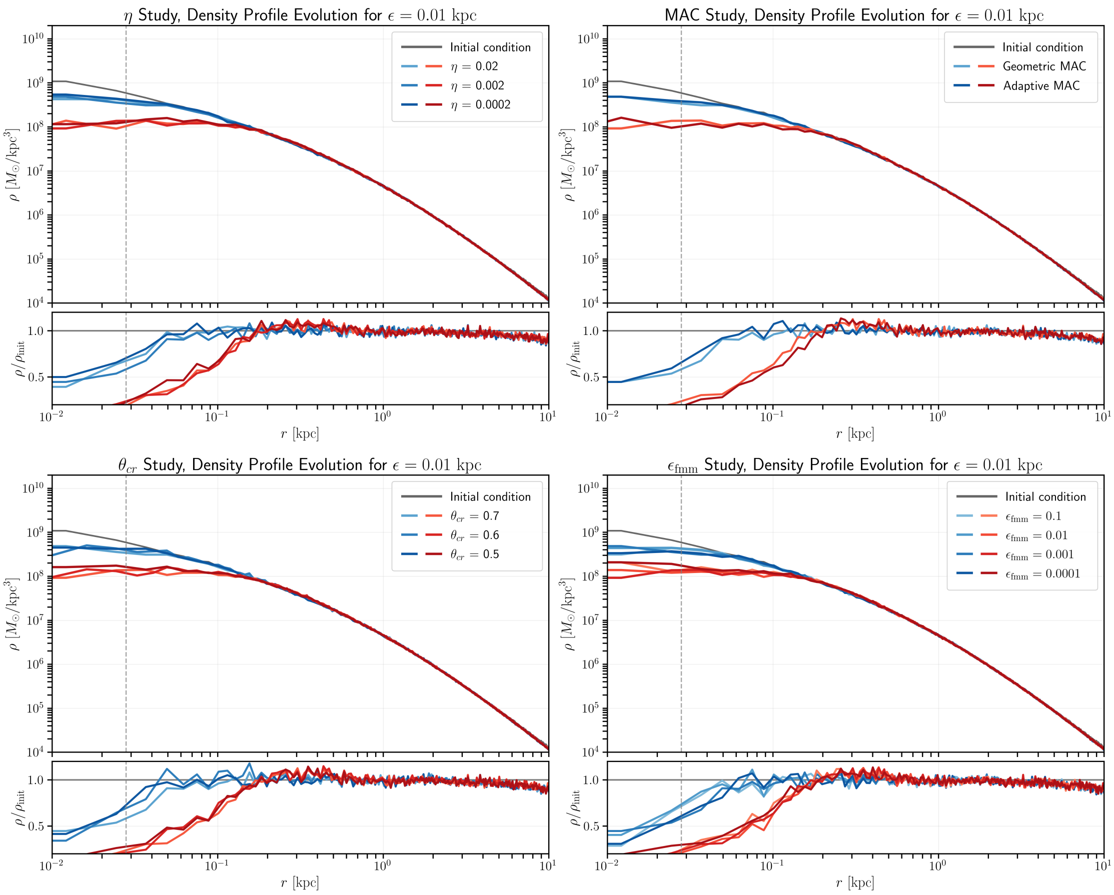

Figure 4.6: **Self-gravity parameter study.** Panels show the central-density evolution for the tested solver settings: top-left – timestep factor η; top-right – MAC choice; bottom-left – geometric opening angle $\theta_{\rm cr}$; bottom-right – FMM tolerance $\epsilon_{\rm fmm}$. In each panel blue curves correspond to 10 Gyr and red curves to 100 Gyr.

The conclusion is that the trend is robust against the standard solver controls. This removes another plausible numerical explanation and strengthens the case for a mechanism tied to the structure of the halo rather than to a single gravity-accuracy parameter.

4.6 Shell study: outer-halo influence on inner potential

The central-density decline observed in the baseline simulation could, in principle, arise from dynamical interactions between the inner halo and the outer shells. To test this hypothesis, we perform an analytic shell-decomposition study using an NFW model (Navarro et al., 1996) to quantify how much of the inner potential depth comes from material at different radii. This analysis then motivates a hybrid numerical experiment: if outer shells are dynamically important, replacing them with an analytic potential while keeping the inner region as particles should produce a measurably different evolution. subsectionAnalytic shell decomposition: method

Following the framework established in Chapter 3, the halo is decomposed into logarithmically spaced shells and the exact analytic contributions of each shell to the gravitational potential and force are computed. The key physical insight is the shell theorem: in a spherically symmetric system, outer shells contribute to the potential everywhere inside them, but only the enclosed mass contributes to the radial force.

For an NFW profile with parameters matching the simulation setup ($\rho_s = 10^7\,M_\odot/\mathrm{kpc}^3$, $r_s = 10\,\mathrm{kpc}$, $r_{\mathrm{vir}} = 100\,\mathrm{kpc}$), the analysis computes: (i) the enclosed mass $M(< r)$ and shell mass distribution, (ii) the contribution of each shell to the potential depth at the center, (iii) the fraction of potential at arbitrary probe radii originating from material outside a chosen cutoff radius, and (iv) a 2D diagnostic map showing these fractions for all radius combinations.

4.6.1 Mass and potential distribution across the halo

The first two diagnostics establish the overall radial structure of the halo. Figure 4.7 shows the enclosed mass profile and the fractional mass distribution per shell.

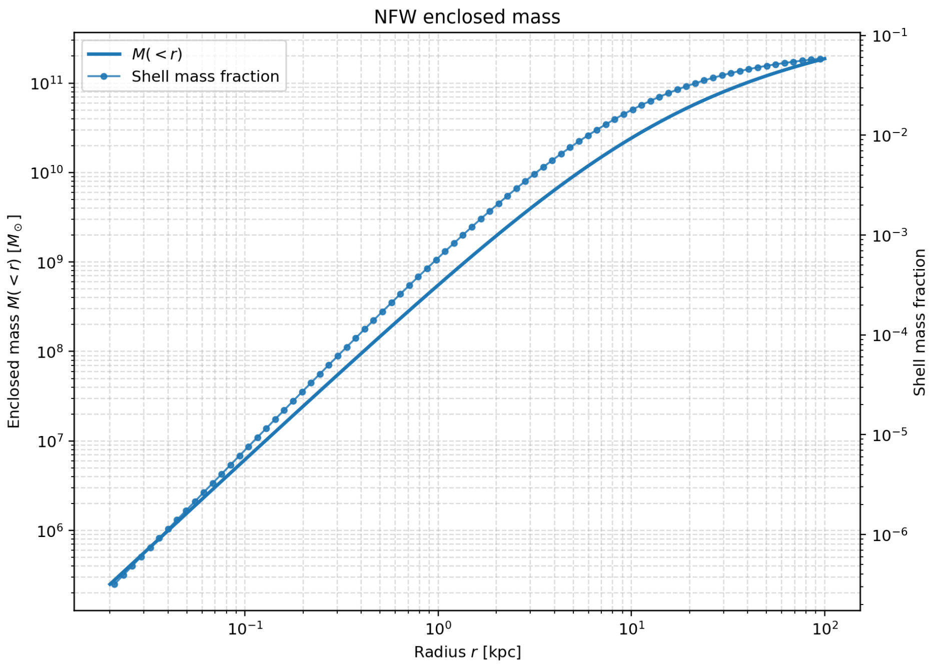

Figure 4.7: **NFW enclosed mass and shell mass fraction.** Left axis: monotonically increasing enclosed mass $M(< r)$ with radius. Right axis: shell mass fraction per logarithmic shell. The outer halo contains a substantial fraction of the total mass budget, indicating that it cannot be ignored when computing potential depths.

The enclosed-mass profile follows the expected NFW form: steep near the center and flattening at larger radii. Importantly, the shell mass fraction plot shows that although the inner shells contain many particles per unit volume (high density), the integrated mass grows progressively outward. Shells at radii $r > 20$ kpc contain a cumulatively significant mass fraction, establishing that the outer halo is not negligible.

Figure 4.8 shows the contribution of each shell to the gravitational potential at the center and the cumulative contribution from all shells outside a given radius.

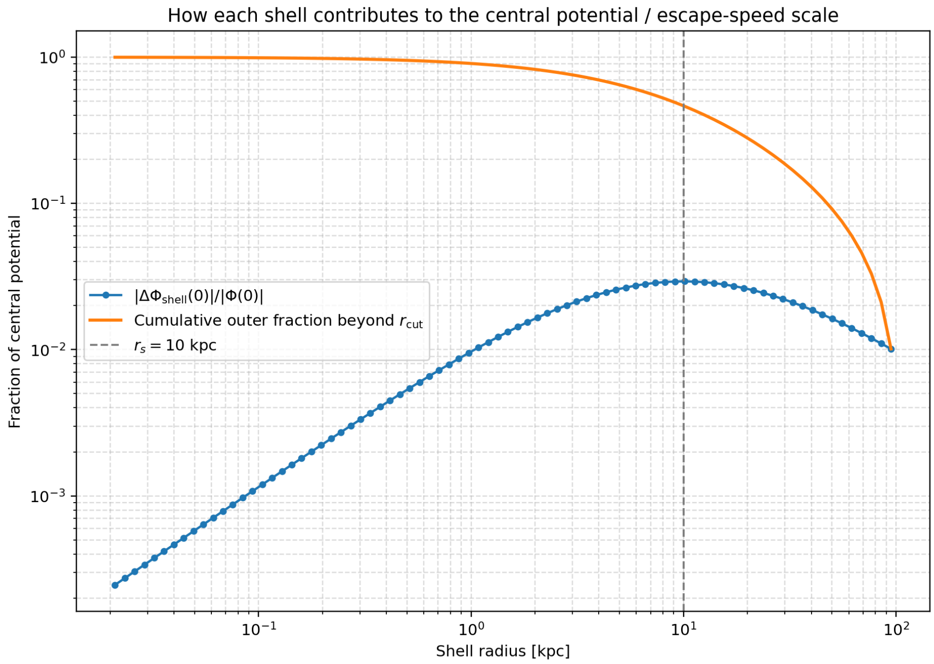

Figure 4.8: **Shell contributions to the central potential depth.** Individual shells (blue points with error bars) contribute to the total potential depth $|\Phi(0)|$. The cumulative contribution (orange line) from all shells beyond a given radius shows that shells at radii beyond the scale radius r_s still make substantial contributions to the escape-speed scale. This supports the argument that the outer halo cannot be treated as a boundary condition; its potential structure must be represented accurately.

The key finding here is that shells well outside the center—at radii several times the scale radius—still contribute visibly to the potential depth. This is exactly the behavior predicted by the shell theorem: outer matter deepens the potential well without changing the local force. The fractional contribution decreases with radius but remains non-negligible up to $r \sim 3r_s$. This result motivates the choice of cutoff radius in the hybrid model: setting r_{cut} too small would omit a significant part of the potential structure.

4.6.2 Outer-halo influence as a function of cutoff radius

The most critical diagnostic for the hybrid model design is the outer-potential fraction: the fraction of the total potential at a given probe radius that arises from material beyond a chosen cutoff radius. Figure 4.9 shows this quantity for several probe radii as the cutoff radius is increased.

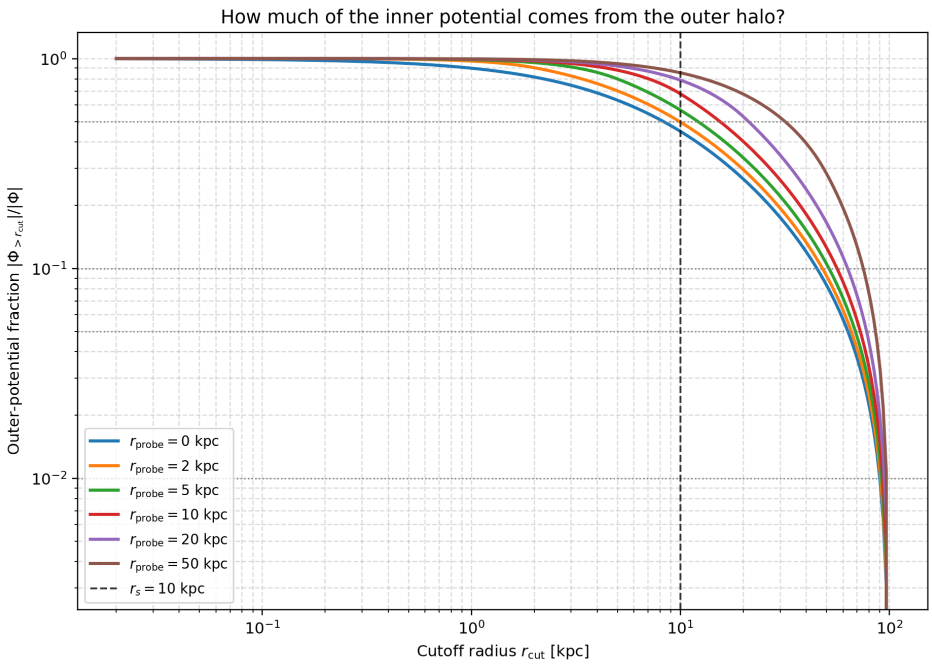

Figure 4.9: **Outer-potential fraction versus cutoff radius.** For probe radii ranging from the center to $r = 5r_s$, the fraction of the potential originating from material beyond the cutoff radius r_{cut} is plotted against the cutoff. As the cutoff moves outward, the outer contribution drops sharply initially, then plateaus. For the central region ($r_{\text{probe}} < r_s$), a cutoff at $r_{\text{cut}} \approx 10$ kpc retains roughly 5–10% of the potential from the outer halo; at $r_{\text{cut}} \approx 30$ kpc, this drops below 1%. This diagnostic directly guides the choice of truncation radius: if the outer halo contributes significantly to the potential at the radii of interest (e.g., $r < 1$ kpc for the central density), then the outer region must be represented accurately.

The curves in this figure answer the central question: how large must r_{cut} be to capture the essential potential structure? For the central region, a cutoff at $r_{\text{cut}} = 10$ kpc (the value used in the hybrid-model simulations) retains the majority of the potential at small radii while excluding the outermost, kinematically less important material. This is a quantitative justification for the parameter choice in the numerical hybrid experiment.

4.6.3 2D outer-fraction map

Figure 4.10 provides a compact summary of the outer-potential fraction across all combinations of probe and cutoff radii.

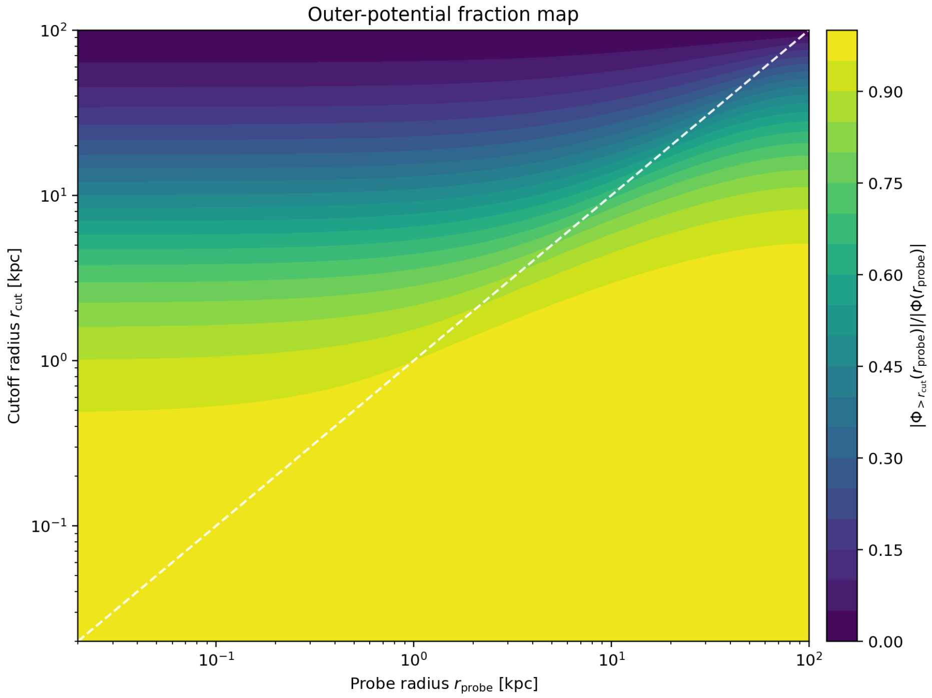

Figure 4.10: **2D map of the outer-potential fraction.** Rows represent probe radii; columns represent cutoff radii. The color intensity indicates the fraction of potential at a given probe radius arising from material beyond the cutoff. The transition from high (red) to low (blue) values shows the radius beyond which the outer halo becomes a small correction. This visualization is useful for selecting practical values of r_{cut} for different regions of interest.

The 2D map reveals the structure clearly: for small cutoff radii (left side), a broad range of inner probe radii (upper rows) feel substantial outer-halo influence. As the cutoff moves outward, this influence shrinks to a narrow band near the cutoff itself. The color gradient provides a quantitative guide for selecting r_{cut} based on acceptable levels of outer-halo contribution.

4.6.4 Interpretation and implications

The analytic shell study establishes that the outer halo is not dynamically negligible: it contributes substantially to the potential structure experienced by inner particles, even if it does not directly modify the local force field through the shell theorem. This has two important consequences:

1. **Motivation for the hybrid model**: A simulation in which the inner region ($r < r_{cut}$)

evolves with particles while the outer region ($r > r_{\mathrm{cut}}$) is represented by an analytic potential can only reproduce the true inner evolution if the outer potential is represented accurately. The shell analysis shows that choosing $r_{\mathrm{cut}} \sim 10$ kpc captures the essential structure while allowing numerical experiments.

2. **Tests of shell-coupling mechanisms**: If the persistent central-density decline arises from global halo coupling—where the outer shells modulate the potential experienced by inner particles—then a hybrid model should produce a noticeably different evolution. Conversely, if the decline persists even in the hybrid setup, the mechanism is either purely inner-region dynamics or depends on structures (e.g., substructure, asymmetries) not captured by a smooth analytic outer potential.

The result is therefore a quantitative prediction: an outer-halo potential that accurately matches the mass distribution should preserve roughly 5–10% of the outer-potential contribution at the cutoff radius. Whether this degree of fidelity is sufficient to maintain the observed central-density decline, or whether further structural complexity is required, is addressed by the hybrid-model simulations discussed in the next section.

4.7 Truncated NFW

This section is not yet completed.

[PLACEHOLDER RESULT] Here we will compare the standard runs against the truncated-NFW hybrid model.

[PLACEHOLDER FIGURE] Baseline versus truncated-NFW central-density evolution.

5 Conclusions

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A An appendix

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